1. Some of the design requirements for the new tires are larger diameter, more tread, and thin. Science and math is applied to these concepts because by adding tires to the chariot with a larger diameter, the same amount of force applied to the chariot will allow the chariot to travel further with one rotation of the wheel because the wheel will be larger. The tires need enough friction to be able to overcome momentum but they can't have too much friction because we don't want the tire to slip, which is why we need more tread and thin tires. Adding more tread will prevent the tire from slipping because it will allow the tire to have more grip, and this means that all of the force being applied will be going into the motion of the chariot. It is also important to have the weight be in the front of the chariot because the chariot is being pulled so the weight needs to be in the front since that is the direction of motion.
2. Science and math is used to justify our final solution because our new tires are more treaded which will allow the chariot to not slip as much. From our discussion with the man at the tire store, tires that are very treaded would be helpful for making turns, but since the chariot runs in a straight line, our new tires are only slightly treaded since that is the best for going straight. Also, the treated tires will be better for the chariot because the chariot needs to run on the wood chips and pebbles in the turf fields.
3. The feedback that we got from the expert at the bike shop was that although we want to change the tires for tires that have a larger diameter, we are unable to do that because of the size of the axle of the chariot. The axle on the chariot tires is much larger than the axle size on most bike tires, so we have to keep the wheel on the tires even though it is slightly dry rotted because it would cost hundreds of dollars to replace the wheel, which is over our budget. The expert at the bike shop also told us that if we want to improve the speed of the chariot, the most effective improvement that we could make would be to get more treaded tires. Additionally, the expert from the bike shop said that the weight in the back of the chariot needs to be as little as possible during the race, which is the same advice we got from the GFS maintenance crew.
4. Our main expert for the tires was the man that worked at the bike shop that we visited. He is qualified to help us because he knows a lot of information and details about the different types and sizes of tires as well as knowing the ways that different types of treads are able to perform.

5. For the same amount of force moving forward, one rotation will travel further.
6. More tread- keeps the tiring from slipping, which means that all of the force is going into the motion of the chariot
7. Thin tire-
8. Want enough friction to overcome momentum but also don't want the tire to slip
9.

- In our conversation with ms.segelken we learned that we need to have more tread in our tires because it will help with the friction and speed. It's the same idea as turf shoes on the turf
- More material = heaver
- Treaded tires = more ridges
- Taller tires= larger diameter (less rotations needed to cover ground)